WEEK 2
WEEKLY PROGRESS REPORT

Synthetic Dataset Preparation and Initial Validation

Project: AI-Based Property Buy-Worthiness Assessment and Property Value Forecasting
Data artifact reviewed: synthetic_properties_raw.csv
Supporting notebook reviewed: minor project analysis.ipynb

1. Summary

During Week 2, the team prepared a synthetic property dataset to support the development of the Property Buy-Worthiness Assessment and Property Value Forecasting prototype while paid real-estate datasets are unavailable. The supplied CSV contains 1,200 records and 32 columns, covering property attributes, locality and access indicators, amenities, legal and builder indicators, civic and utility fields, and past five-year price growth.

Initial inspection found 36 locality categories and four property types. There are no exact duplicate rows and no duplicate property IDs.The dataset is suitable as a starting point for software prototyping and controlled experiments, subject to the data-quality actions and file/column alignment issues documented in this report. It must not be described as verified market data or used to claim real-world forecasting accuracy.

2. Week 2 Objectives

- Prepare a structured dataset for the planned property assessment and forecasting workflow.
- Represent property, location, amenities, legal/builder, civic/utility and market-history information.
- Check basic integrity: dataset dimensions, identifiers, missing values, locality coverage and price arithmetic.
- Record the dataset's synthetic status and assumptions so later results can be interpreted correctly.
- Prepare the dataset and documentation needed for Week 3 preprocessing and exploratory data analysis (EDA).

3. Dataset Overview

Item	Observed result
Input file	synthetic_properties_raw.csv
Rows / property records	1,200
Columns in supplied CSV	32
Unique property IDs	1,200
Unique locality labels	36
Property types	4
Missing cells	78 across 5 columns
Exact duplicate rows	0
Duplicate property IDs	0
Synthetic-status field	All 1,200 records
Generation seed field	One value present: 20261013

13. Conclusion

Week 2 has produced a structured synthetic dataset of 1,200 property records across 36 simulated locality labels. Basic integrity checks show unique property IDs, no duplicate rows and consistent asking-price arithmetic, while 78 missing cells require documented handling. The main outstanding handover issue is ensuring that the notebook, raw CSV, recommendation labels and future-value target columns are aligned. Once this is resolved, the dataset can proceed to Week 3 preprocessing and EDA.